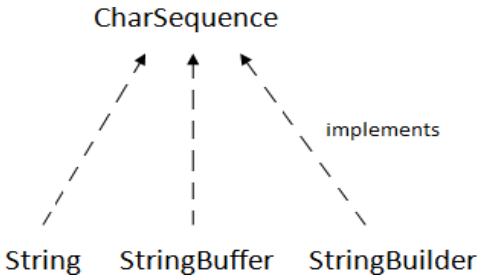
Strings in Java:

Definition:- Strings are sequences of Unicode characters. In many programming languages strings are stored in *arrays* of characters. However, in Java strings are a separate object type, String. The "+" operator is used for concatenation, but all other operations on strings are done with methods in the String class.

The CharSequence interface is used to represent sequence of characters. It is implemented by String, StringBuffer and StringBuilder classes. It means, we can create string in java by using these 3 classes.



The java String is immutable i.e. it cannot be changed. Whenever we change any string, a new instance is created. For mutable string, you can use StringBuffer and StringBuilder classes.

Java provides 3 types of string handling classes

String class -defined in "java.lang" package

- Once an object is created, no modifications can be done on that.

StringBuffer class and StringBuilder class– defined in "java.lang" package

- Modifications can be allowed on created object.

StringTokenizer class - defined in "java.util" package

- Used to divide the string into substrings based on tokens.

Declaration & Creation of String:

Syntax: String string_name; //Decl

String_name=new String("string"); //creation

Ex: String sname;

sname=new String(" Java");

(or)

String sname=new String("java"); //Both

Java Strings can be concatenated using the '+' operator Ex: String Fullname=Name1+Name2;

If Name1="latha",Name2="Reddy", we get Fullname="LathaReddy"

String Arrays:

Java also supports the cration & usage of arrays that contain strings

Syntax: String str_array_name[ ]=new

String[size]; Ex: String
ItemArray[ ]=new String[6];

String Constructors:-

1. String s= new String(); //To create an empty String call the default constructor. 2. char chars*+=,‘a’,’b’,’c’,’d’-;
   String s=new String(chars);
3. String(String str); //Construct a string object by passing another string object. String str = "abcd";
   String str2 = new String(str);
4. String(char chars[], int startIndex, int numChars)
   //To create a string by specifying positions from an array of characters char chars*+=,‘a’,’b’,’c’,’d’,’e’,’f’-;
   String s=new String(chars,2,3); //This initializes **s** with characters “cde”.
5. String(byte asciiChars[ ]) // Construct a string from subset of byte array.
6. String(byte asciiChars[ ], int startIndex, int numChars)

Examples:-

```
// Construct one String from another.
                class MakeString {
public static void main(String args[])
 { char c[] = {'J', 'a', 'v', 'a'};
String s1 = new String(c);
String s2 = new String(s1);
 System.out.println(s1);
System.out.println(s2);
                        }
                }
```

Outpu
t: Java
Java

```
// Construct string from subset of char array.
                class SubStringCons {
                        public static void main(String
                           args[]) { byte ascii[] = {65,
                           66, 67, 68, 69, 70 };
                                String s1 = new
                                String(ascii);
                                System.out.println(s1)
                                ;
                                String s2 = new
                                String(ascii, 2, 3);
                                System.out.println(s2);
                        }
                }
```

Outpu
t
ABCD
EF
CDE

String Methods:-

Method Call	Retu rn Type	Task Performed
s2=s1.toLowerCase()	String	Converts the string s1 to all lowercase
s2=s1.toUpperCase()	String	Converts the String s1 to all Uppercase
s2=s1.replace('x','y')	String	Replaces all appearances of 'x' with 'y'
s2=s1.trim()	String	Remove white spaces at the beginning & end of the string s1
s1.equals(s2)	Boolean	Returns 'true' if s1 is same as s2, same characters in same order (case-sensitive) 'false' otherwise
s1.equalsIgnoreCase(s2)	Boolean	'' '' , ignoring the case of characters
s1.length()	Int	Returns the no. of characters in s1
s1.charAt(n)	char	Returns the nth characters in s1
s1.getChars(m,n,c,0)	Void	Extracts more than one character at a time & copies into a character array 'c' m-source start n- source end 0-Target start
s1.compareTo(s2)	Int	Returns zero, if s1=s2 (case sensitive) -ve , if s1<s2 +ve , if s1>s2
s1.compareToIgnoreCase(s2)	Int	ignoring case sensitive
s1.regionMatches(m,s2,	Boolean	Compares a specific region inside a

n,c) s1.regionMatches(B,m,s2,n,c)	 Boolean	string with another specific region in another string True if same Else False m-start index of s1 n-start index of s2 c-no. of characters to compare if B is true ,ignores case
s1.indexOf('ch') s1.indexOf("str")	Int	Searches for the 1st occurrence of a character or string in s1 & returns the index value else returns '- 1'
s1.indexOf('ch',n) s1.indexOf("str",n)	Int	'' '' from the nth position in s1
s1.lastIndexOf('ch') s1.lastIndexOf("str")	Int	Searches for last occurrence of a character or string in s1. The search is performed from the end of the string towards beginning & returns the index value else '-1'
S1.lastIndexOf('ch',n) S1.lastIndexOf("str",n)	Int	'' '' from the nth position from end to begin
S1.substring(n)	String	Gives substring starting from nth character
S1.substring(n,m)	String	'' '' '' upto mth character (not including mth character)
S1.startsWith("str")	Boolean	Determines whether a given string 's1' begins with a specific string or not
S1.endsWith("str")	Boolean	'' '' '' '' ends '' ''
S1.concat(s2)	String	Concatenates s1&s2 stored

S1.toCharArray()	Char [ ]	Returns a character array containing a copy of the characters in a string
o.toString()	String	Converts all objects into string objects
String.valueOf(o)	String	Creates a string object of the parameter ‘o’ (Simple type or Object type)
String.valueOf(var)	String	Converts the parameter value to String representation

Character Extratction:-

```
class getCharsDemo {
public static void main(String args[]) {
        String s = "This is a demo of the
                                                getChars
        method."; int start = 10;
        int end = 14;
        char buf[] = new char[end
        - start]; s.getChars(start,
        end, buf, 0);
        System.out.println (buf);
        }
}
```

Outp ut:

demo

```
class GetBytesDemo{
        public static void
          main(String[] args){ String
          str = "abc“ + ”ABC”;
                byte[] b = str.getBytes();
                //char[]
                c=str.toCharArray();
                System.out.println(str)
                ; for(int
                i=0;i<b.length;i++){
                  System.out.print(b[i]+" ");
                  //System.out.print(c[i]+"  ");
                }
        }
}
```

Output:

97 98 99 65 66 67

//a b c A B C

String Comparison:

1. boolean **equals**(Object str)
   To compare two strings for equality. It returns true if the strings contain the same characters in the same order, and false otherwise.
2. boolean **equalsIgnoreCase**(String str)
   To perform a comparison that ignores case differences.

```
// Demonstrate equals() and
            equalsIgnoreCase(). class
            equalsDemo {
                  public static void main(String args[]) {
```

```
                String s1 =
                "Hello"; String s2
                = "Hello"; String
                s3 = "Good-bye";
                String s4 =
                "HELLO";
                System.out.println(s1.equals(s2));
                System.out.println(s1.equals(s3));
                System.out.println(s1.equals(s4));
                System.out.println(s1.equalsIgnoreCase(s4));
        }
}
```

String Comparison using regionMatches Method:-

boolean **regionMatches**(int startIndex, String str2, int str2StartIndex, int numChars)

boolean **regionMatches**(boolean ignoreCase, int startIndex, String str2, int str2StartIndex, int numChars)

- The regionMatches( ) method compares a specific region inside a string with another specific region in another string.
- startIndex specifies the index at which the region begins within the invoking String object.
- The String being compared is specified by str2. The index at which the comparison will start within str2 is specified by str2StartIndex.
- The length of the substring being compared is passed in numChars.

Example:-

```
class RegionTest{
        public static void main(String
                args[]){ String str1 =
                "This is Test"; String
                str2 = "THIS IS TEST";
                if(str1.regionMatches(5,str2,5,3)) {
                        // Case,
                        pos1,secdString,pos1,len
                        System.out.println("Strings are
                        Equal");
                 }
                els
                e{      System.out.println("Strings are NOT Equal");

                }
        }
```

```
        }
```

Output:

Strings are NOT Equal

```
//Sorting of String using compareTo
        method. class SortString
        {
            static String arr[] = { "is", "the", "time", "for", "all", "good",
            "men", "to",
        "come", "to", "the", "aid", "of", "their",
            "country"-; public static void
            main(String args[]) {
                for(int j = 0; j < arr.length; j++) {
                    for(int i = j + 1; i < arr.length;
                        i++)
                        { if(arr[i].compareTo(ar
                        r[j]) < 0) {
                            String t =
                            arr[j]; arr[j]
                            = arr[i];
                            arr[i] = t;
                        }
                    }
                    System.out.println(arr[j]);
                }
            }
        }
```

Changing the Case of Characters Within a String :-
String toLowerCase( ):- converts all the characters in a string from uppercase to lowercase. String toUpperCase( ):- converts all the characters in a string from lowercase to uppercase

```
// Demonstrate toUpperCase() and toLowerCase().
        class ChangeCase {
            public static void main(String
                args[]){ String s = "This is a
                test.";
                System.out.println("Original
                : " + s); String upper =
                s.toUpperCase(); String
                lower = s.toLowerCase();
                System.out.println("Uppercase: "
                + upper);
                System.out.println("Lowercase: "
                + lower);
            }
        }
```

Output:

```
Original: This is a
test. Uppercase: THIS
IS A TEST.
Lowercase: this is a
test.
```

StringBuffer Class:

- StringBuffer is mutable means one can change the value of the object .
- The object created through StringBuffer is stored in the heap. StringBuffer has the same methods as the StringBuilder , but each method in StringBuffer is synchronized that is StringBuffer is thread safe . Due to this it does not allow two threads to simultaneously access the same method . Each method can be accessed by one thread at a time .
- StringBuffer is a peer class of String that provides much of the functionality of strings. String represents fixed-length, immutable character sequences while StringBuffer represents growable and writable character sequences.

StringBuffer Constructors

1. StringBuffer( ): It reserves room for 16 characters without reallocation. StringBuffer s=new StringBuffer();
2. StringBuffer(int size)It accepts an integer argument that explicitly sets the size of the buffer. StringBuffer s=new StringBuffer(20);
3. StringBuffer(String str): It accepts a String argument that sets the initial contents of the StringBuffer object and reserves room for 16 more characters without reallocation. StringBuffer s=new

StringBuffer("RGUKTIIIT");

Methods

1. StringBuffer append() method

   The append() method concatenates the given argument with this string.

```
class StringBufferExample{
public static void main(String
args[]){ StringBuffer sb=new
StringBuffer("Hello ");
sb.append("Java");//now original string is
changed System.out.println(sb);//prints
Hello Java
}
}
```

2. StringBuffer insert() method

   The insert() method inserts the given string with this string at the given position.

```
class StringBufferExample2{
public static void main(String
args[]){ StringBuffer sb=new
StringBuffer("Hello ");
sb.insert(1,"Java");//now original string is
changed System.out.println(sb);//prints
HJavaello
}
}
```

3. StringBuffer replace() method

   The replace() method replaces the given string from the specified beginIndex and endIndex.

```
class StringBufferExample3{
public static void main(String
args[]){ StringBuffer sb=new
StringBuffer("Hello");
sb.replace(1,3,"Java");
System.out.println(sb);//prints
HJavalo
}
}
```

4. StringBuffer delete() method

   The delete() method of StringBuffer class deletes the string from the specified beginIndex to endIndex.

```
class StringBufferExample4{
public static void main(String
args[]){ StringBuffer sb=new
StringBuffer("Hello");
sb.delete(1,3);
System.out.println(sb);//prints Hlo
}
}
```

5. StringBuffer reverse() method
The reverse() method of StringBuilder class reverses the current string.

```
class StringBufferExample5{
public static void main(String
args[]){ StringBuffer sb=new
StringBuffer("Hello"); sb.reverse();
System.out.println(sb);//prints
olleH
}
}
```

6. StringBuffer capacity() method
The capacity() method of StringBuffer class returns the current capacity of the buffer. The default capacity of the buffer is 16. If the number of character increases from its current capacity, it increases the capacity by (oldcapacity*2)+2. For example if your current capacity is 16, it will be (16*2)+2=34.

```
class StringBufferExample6{
public static void main(String args[]){
StringBuffer sb=new StringBuffer();
System.out.println(sb.capacity());//d
efault 16 sb.append("Hello");
System.out.println(sb.capacity());//n
ow 16 sb.append("java is my
favourite language");
System.out.println(sb.capacity());//now (16*2)+2=34 i.e
(oldcapacity*2)+2
}
}
```

7. StringBuffer ensureCapacity() method
The ensureCapacity() method of StringBuffer class ensures that the given capacity is the minimum to the current capacity. If it is greater than the current capacity, it increases the capacity by (oldcapacity*2)+2. For example if your current capacity is 16, it will be (16*2)+2=34.

```
class StringBufferExample7{
public static void main(String
args[]){ StringBuffer sb=new
StringBuffer();
System.out.println(sb.capacity());//d
efault 16 sb.append("Hello");
System.out.println(sb.capacity());//n
ow 16 sb.append("java is my
favourite language");
System.out.println(sb.capacity());//now (16*2)+2=34 i.e
(oldcapacity*2)+2 sb.ensureCapacity(10);//now no change
System.out.println(sb.capacity());//now 34
sb.ensureCapacity(50);//now (34*2)+2
System.out.println(sb.capacity());//now 70
}
}
```